

Projects, Papers, and Proteins

Do student iGEM projects predict local academic research in synthetic biology?

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June 2026

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Motivation

iGEM is a global natural experiment in distributed student innovation

- 100k+ participants; 400+ universities; 50+ countries
- Teams design organisms around a local problem
- Projects, parts, and wiki pages are publicly archived
- iGEM runs annually – a repeated cross-section from 2009 to 2025

The iGEM corpus gives us a rare window into how student-level scientific interest maps onto local research infrastructure.

The question: are student projects topically embedded in local research?

Inspired by:

- Hidalgo et al. (2007): revealed comparative advantage and product space
- Neffke & Henning (2013): skill-relatedness from labor flows

Applied here to text, not products or occupations.

- Unit: city
- Similarity: cosine distance in embedding space
- Question: do student projects and local papers occupy related regions of semantic space?

Core hypothesis

Innovation follows a local path:
student projects → papers → patents
If so: project topics should *lead* paper topics in the same city by a few years.

Data and Measurement

Three corpora, one embedding space

Corpus	Docs	Cities	Years	Source
Synbio papers	10,402	1,218	1964–2026	OpenAlex
iGEM projects	4,614	626	2009–2025	iGEM Registry
BioBrick parts	89,060	472	2009–2025	iGEM Registry
Analysis sample	–	387	–	cities with both

Carbon-capture tagged: 294 papers (2.8%) and 141 projects (3.1%).

SPECTER2 embeds scientific text into a citation-informed space

- 768-dim vectors; contrastive learning on citation pairs
- **Proximity adapter**: fine-tuned for semantic similarity, not classification
- Cached for all 15,000 documents; fully reproducible

City representation:

$$\bar{\mathbf{a}}_c = \frac{1}{|P_c|} \sum_{p \in P_c} \mathbf{e}_p$$

Mean embedding over all docs of type t in city c .

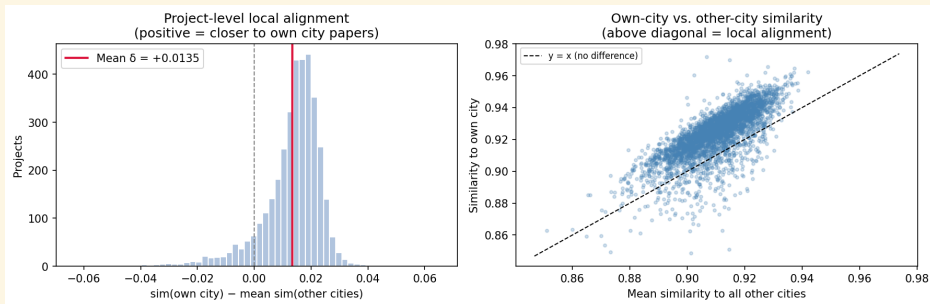
Why SPECTER2?

Standard sentence transformers are not calibrated for scientific text.

SPECTER2 is trained on 146M citation pairs; documents that co-cite tend to be nearby.

Result 1: Local Alignment

90% of projects are semantically closer to their own city's papers



$n = 3,620$ projects; cities with ≥ 3 papers; $\delta = \text{own-city cosine} - \text{mean over all other cities}$

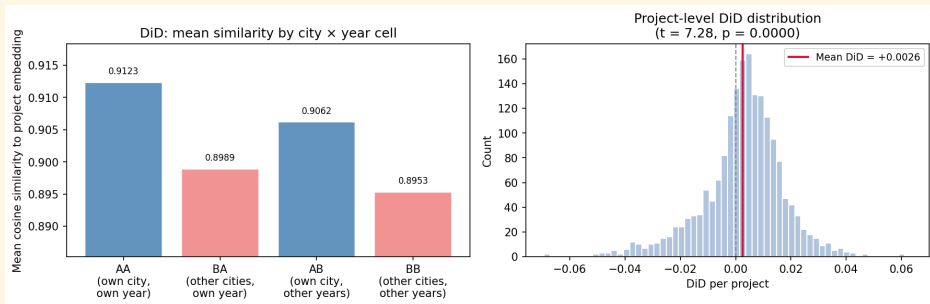
Local alignment: robust across two tests

	Statistic	<i>p</i> -value
Mean δ (own – other)	+0.0135	
Fraction with $\delta > 0$	90.6%	
One-sample <i>t</i> -test	$t = 78.02$	< 0.001
Wilcoxon signed-rank		< 0.001

Takeaway

Projects are systematically closer to local papers than to papers anywhere else. The effect is small but extremely consistent.

DiD: separating stable specialisation from temporal alignment



Annual city-year centroids (≥ 2 papers); 1,841 projects in DiD sample

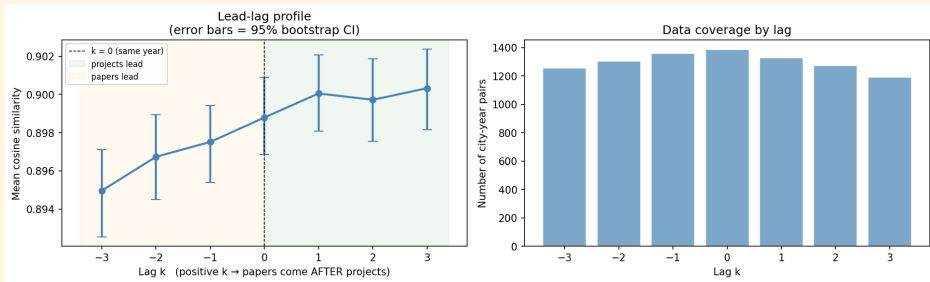
The temporal component is real but modest

	Own city	Other cities	Local advantage
Own year	0.9123	0.8989	+0.0135
Other years	0.9062	0.8953	+0.0109
Δ (temporal)	+0.0062	+0.0036	
DiD		+0.0026	

Stable city specialisation accounts for most of the alignment. The DiD component is small but positive.

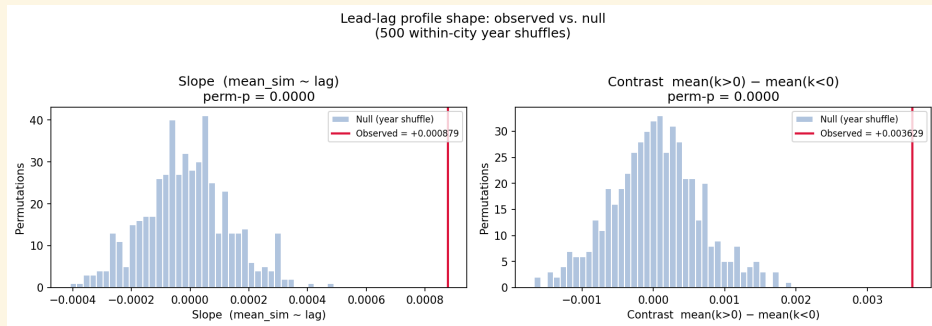
Result 2: Lead-Lag Structure

Projects are semantically ahead of local papers by about 3 years



Lag k : cosine similarity between project centroid in year t and paper centroid in year $t + k$; bootstrapped 95% CI

The lead-lag slope is not a temporal sampling artifact



500 within-city year-shuffles; observed slope = +0.000879; null $p_{95} = +0.000246$; perm- $p < 0.05$

A Measurement Problem

Centroid cosine similarity is dominated by document count

Spearman correlations with semantic overlap:

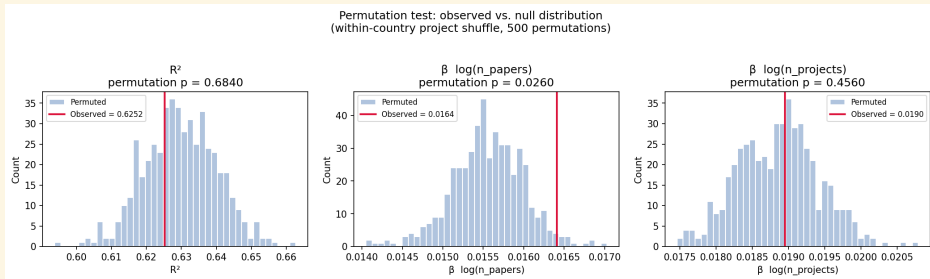
Variable	ρ
$\log n_{\text{papers}}$	0.78
$\log n_{\text{projects}}$	0.83

OLS $R^2 = 0.625$ with only two log-count variables.

The problem

Larger cities have more stable centroids. Two stable centroids drawn from *any* synthetic biology corpus will be similar – because the field occupies a compact region of embedding space. Overlap ≈ 0.946 everywhere.

Cross-section permutation: centroid overlap fails



500 within-country shuffles of project city labels; OLS on shuffled data

Permutation results: centroid similarity is not locally specific

Statistic	Observed	Null mean	Null p_{95}	perm- p
R^2	0.625	0.630	0.646	0.684
$\beta \log n_{\text{papers}}$	0.0164	0.0156	0.0163	0.026
$\beta \log n_{\text{projects}}$	0.0190	0.0189	0.0198	0.456

Takeaway

The centroid-cosine measure does not distinguish real local alignment from random assignment. We need a better measure of specificity.

Result 3: Cluster Co-Membership

UMAP + HDBSCAN finds 83 topic clusters across all 15k documents

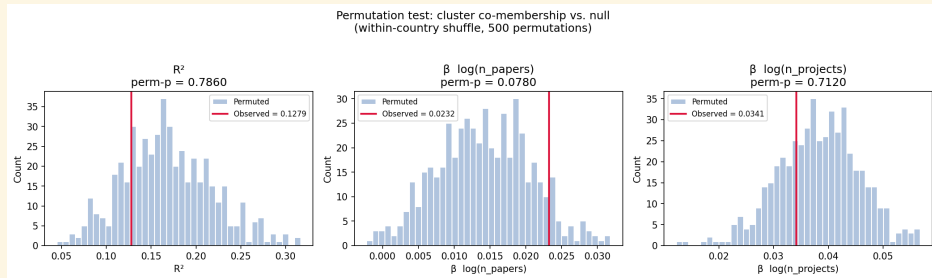
- UMAP: 60 dimensions, cosine metric
- HDBSCAN: 83 clusters, $\sim 20\%$ noise
- Cluster overlap = weighted Jaccard on cluster frequency vectors
- Sample: 189 cities with ≥ 3 clustered docs per type

Mean cluster overlap: 0.081; median: 0.023;
max: 0.717

Why this is better

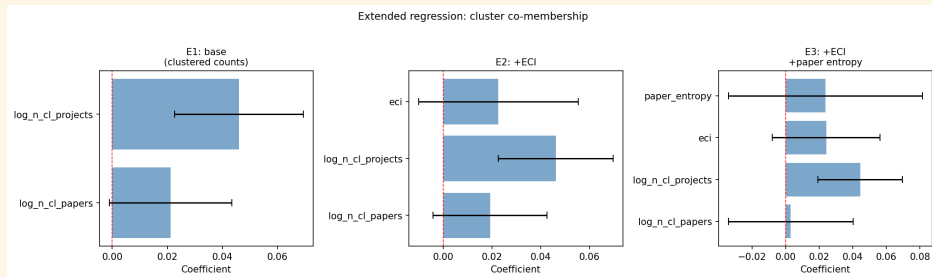
Cluster co-membership is *relative*. It asks: do a city's projects and papers concentrate in the *same topical niches*, not just the same broad field?

Cluster co-membership survives the permutation test



500 within-country shuffles; observed $\beta \log n_{\text{projects}} = +0.046^{***}$; perm- $p = 0.03$

Coefficient plot: cluster co-membership is robust



OLS with country FE and robust SE; dependent variable: cluster overlap score

Result 4: Cross-Modal Validation

Physical part-type composition independently agrees with semantic clusters

BioBrick parts: standardised DNA sequences registered by teams

- Types: promoter, CDS, RBS, terminator, reporter, device, ...
- City part-type profile: proportion vector over part types
- Shannon entropy measures specialisation

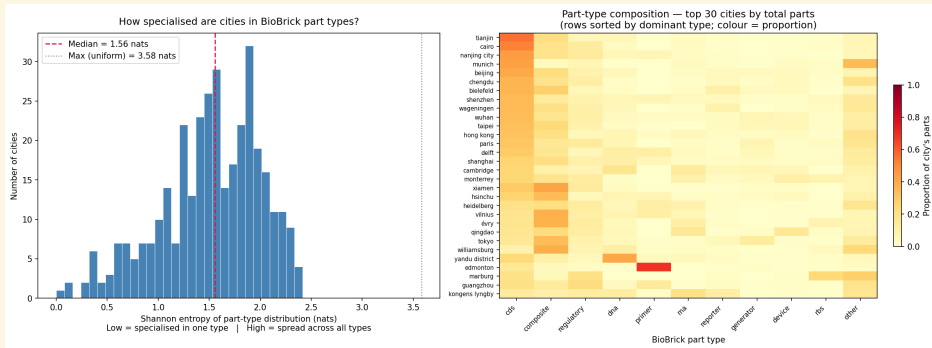
Mantel test: are part-type distances correlated with cluster-space distances?

Result (179 cities, 999 permutations)

$$r = 0.2207, \quad p < 0.001$$

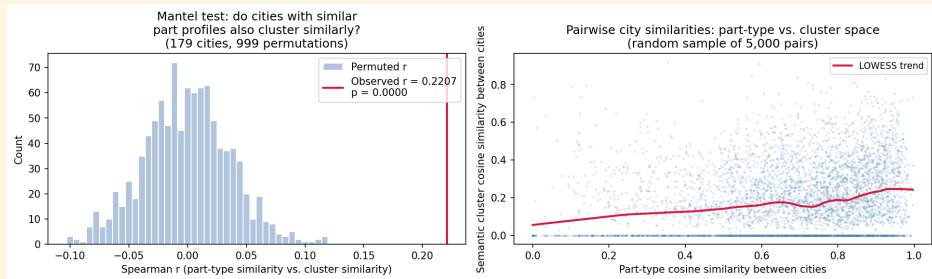
Null $p_{95} = 0.073$. Part-type space and semantic cluster space agree.

Part-type specialisation varies meaningfully across cities



Top 30 cities by total registered parts; sorted by dominant part type

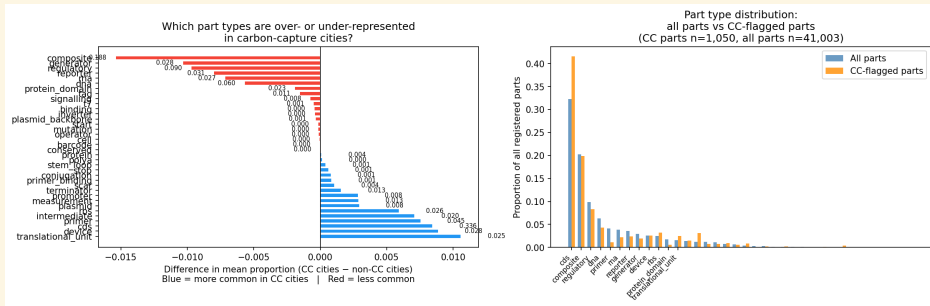
Mantel test: part-type and cluster distances are correlated



Spearman Mantel $r = 0.2207$; perm- $p < 0.001$; 179 cities

Case Study: Carbon Capture

Carbon capture teams have a distinctive part-type signature



71 CC cities vs. 277 non-CC cities; CC = at least one carbon-capture-flagged team

Carbon capture: what we see and what we cannot yet say

Consistent with the hypothesis:

- CC teams deposit more CDS parts (enzyme-encoding sequences)
- CC cities have above-average cluster co-membership
- CC papers and projects share semantic clusters

What this is not:

- Not evidence of causal spillover
- CC flag is based on keyword retrieval, not expert review
- Small n limits inference on CC-specific lead-lag

Summary

Four results, in order of strength

1. **Local alignment is real and consistent** – 90.6% of projects are topically closer to their own city's papers ($t = 78$, $p < 0.001$).
2. **Projects lead papers by ≈ 3 years** – lead-lag peak at $k = +3$; profile shape non-random under permutation.
3. **Cluster co-membership is locally specific** – survives within-country permutation where centroid similarity does not ($p = 0.03$).
4. **Part-type space corroborates semantic space** – Mantel $r = 0.22$, $p < 0.001$; two independent signals agree.

Limitations and what comes next

Current limitations:

- No causal identification – shared local environment could drive both
- Geocoding errors propagate through all city-level results
- HDBSCAN labels are not stable across hyperparameter choices
- Patents not yet integrated

Next steps:

- Add patent corpus (Lens.org)
- Instrument for iGEM presence using competition history
- Quarto site: interactive semantic space explorer + geographic view
- Manuscript draft for submission

The main claim, stated carefully

In cities with active iGEM participation, the topics that student teams work on are semantically related to the topics that local researchers publish on – and this co-location in semantic space is not explained by city size alone, holds at the topic-cluster level, and is corroborated by an independent physical signal (BioBrick part-type composition). Whether this reflects knowledge spillovers, shared local opportunity, or selection of students into research-active cities remains open.